

FDAHW3

```
# =====  
# GM weekly log-returns ~ Toyota + Ford  
# =====  
  
dat <- read.table("w_logret_3automanu.txt", header = TRUE, sep = "\t",  
                 stringsAsFactors = FALSE)  
dat$Date <- as.Date(dat$Date, format = "%m/%d/%Y")  
  
cat("=== Data summary ===\n")
```

```
## === Data summary ===
```

```
str(dat)
```

```
## 'data.frame': 709 obs. of 4 variables:  
## $ Date : Date, format: "1994-01-03" "1994-01-10" ...  
## $ Toyota: num 0.01343 0.00491 -0.00332 0.01925 0.0108 ...  
## $ Ford : num 0.000586 0.023912 -0.005014 -0.01193 0.017498 ...  
## $ GM : num 0.02307 0.02471 -0.00266 -0.01187 0.01718 ...
```

```
print(summary(dat[, c("Toyota", "Ford", "GM")]))
```

```
##      Toyota      Ford      GM  
## Min.   :-0.1132272 Min.   :-9.481e-02 Min.   :-0.1061545  
## 1st Qu.: -0.0084958 1st Qu.: -9.368e-03 1st Qu.: -0.0119514  
## Median : 0.0005180 Median : -2.082e-04 Median : 0.0004089  
## Mean   : 0.0008693 Mean    : 7.535e-05 Mean    : 0.0001701  
## 3rd Qu.: 0.0105164 3rd Qu.: 1.122e-02 3rd Qu.: 0.0116914  
## Max.   : 0.1049635 Max.    : 7.429e-02 Max.    : 0.0744415
```

```
cat("\nN obs:", nrow(dat), "\n\n")
```

```
##  
## N obs: 709
```

```
# ---- Fit model ----  
fit <- lm(GM ~ Toyota + Ford, data = dat)  
  
cat("\n=== Model summary ===\n")
```

```
##  
## === Model summary ===
```

```
print(summary(fit))
```

```
##
## Call:
## lm(formula = GM ~ Toyota + Ford, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.062848 -0.009649 -0.000405  0.008977  0.057515
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 7.049e-05  5.914e-04   0.119   0.905
## Toyota      6.132e-02  3.784e-02   1.621   0.106
## Ford        6.145e-01  3.132e-02  19.619 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.01572 on 706 degrees of freedom
## Multiple R-squared:  0.3775, Adjusted R-squared:  0.3757
## F-statistic: 214.1 on 2 and 706 DF, p-value: < 2.2e-16
```

```
cat("\n=== ANOVA ===\n")
```

```
##
## === ANOVA ===
```

```
print(anova(fit))
```

```
## Analysis of Variance Table
##
## Response: GM
##           Df Sum Sq Mean Sq F value    Pr(>F)
## Toyota     1 0.010682  0.010682   43.21 9.559e-11 ***
## Ford       1 0.095151  0.095151  384.90 < 2.2e-16 ***
## Residuals 706 0.174532  0.000247
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
cat("\n=== Confidence intervals (95%) ===\n")
```

```
##
## === Confidence intervals (95%) ===
```

```
print(confint(fit))
```

```
##              2.5 %      97.5 %
## (Intercept) -0.001090582 0.001231555
## Toyota      -0.012971455 0.135614228
## Ford         0.553001107 0.675991088
```

```

# ---- Diagnostics ----
cat("\n=== VIF (multicollinearity) ===\n")

##
## === VIF (multicollinearity) ===

# manual VIF (avoid needing 'car' package)
vif_manual <- function(model) {
  mm <- model.matrix(model)[, -1, drop = FALSE]
  vifs <- sapply(colnames(mm), function(cn) {
    others <- setdiff(colnames(mm), cn)
    r2 <- summary(lm(mm[, cn] ~ mm[, others]))$r.squared
    1 / (1 - r2)
  })
  vifs
}
print(vif_manual(fit))

##   Toyota   Ford
## 1.062458 1.062458

cat("\n=== Durbin-Watson (autocorrelation) - manual calc ===\n")

##
## === Durbin-Watson (autocorrelation) - manual calc ===

res <- resid(fit)
dw <- sum(diff(res)^2) / sum(res^2)
cat("DW statistic:", round(dw, 4), "(2 = no autocorrelation)\n")

## DW statistic: 2.0158 (2 = no autocorrelation)

cat("\n=== Breusch-Pagan test (heteroskedasticity) - manual ===\n")

##
## === Breusch-Pagan test (heteroskedasticity) - manual ===

bp_aux <- lm(res^2 ~ dat$Toyota + dat$Ford)
bp_stat <- nrow(dat) * summary(bp_aux)$r.squared
bp_p <- 1 - pchisq(bp_stat, df = 2)
cat("BP statistic:", round(bp_stat, 4), " p-value:", round(bp_p, 4), "\n")

## BP statistic: 4.6929 p-value: 0.0957

cat("\n=== Shapiro-Wilk normality test on residuals ===\n")

##
## === Shapiro-Wilk normality test on residuals ===

```

```
print(shapiro.test(res))
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: res  
## W = 0.98551, p-value = 1.763e-06
```

```
cat("\n=== Influential points (Cook's distance > 4/n) ===\n")
```

```
##  
## === Influential points (Cook's distance > 4/n) ===
```

```
cooksds <- cooks.distance(fit)  
thresh <- 4 / nrow(dat)  
infl_idx <- which(cooksds > thresh)  
cat("Number flagged:", length(infl_idx), "out of", nrow(dat), "\n")
```

```
## Number flagged: 55 out of 709
```

```
print(head(dat[infl_idx, c("Date", "Toyota", "Ford", "GM")][order(-cooksds[infl_idx]),], 10))
```

```
##           Date      Toyota      Ford      GM  
## 402 2001-09-10 -0.113227212 -0.087458033 -0.106154533  
## 644 2006-05-08 -0.043508275  0.011216263  0.055571928  
## 311 1999-12-13  0.104963521  0.009880111 -0.010523804  
## 334 2000-05-22 -0.016471746 -0.044885538 -0.091369322  
## 624 2005-12-19  0.023229841 -0.021622187 -0.065486975  
## 629 2006-01-23 -0.000508442  0.044790052  0.074441495  
## 584 2005-03-14 -0.012760108 -0.036667800 -0.079237179  
## 484 2003-04-14  0.016158265  0.064856319  0.006644412  
## 413 2001-12-03 -0.011425965 -0.043708928  0.016189517  
## 315 2000-01-10  0.008454716 -0.023563980  0.041151923
```

```
# ---- Save diagnostic plots ----  
png("diagnostics.png", width = 1000, height = 1000, res = 120)  
par(mfrow = c(2,2))  
plot(fit)  
dev.off()
```

```
## pdf  
## 2
```

```
png("scatter.png", width = 1100, height = 500, res = 120)  
par(mfrow = c(1,2))  
plot(dat$Toyota, dat$GM, xlab = "Toyota weekly log-return", ylab = "GM weekly log-return",  
     main = "GM vs Toyota", pch = 20, col = rgb(0,0,0,0.4))  
abline(lm(GM ~ Toyota, data = dat), col = "red", lwd = 2)  
plot(dat$Ford, dat$GM, xlab = "Ford weekly log-return", ylab = "GM weekly log-return",  
     main = "GM vs Ford", pch = 20, col = rgb(0,0,0,0.4))  
abline(lm(GM ~ Ford, data = dat), col = "red", lwd = 2)  
dev.off()
```

```
## pdf
## 2
```

```
cat("\nDone.\n")
```

```
##
## Done.
```